

Problem 1

Using the training and test sets specified in the syllabus, perform the following tasks:

1. On the Gisette dataset, train decision trees of maximum depth $1, 2, \dots$ up to 12, for a total of 12 decision trees. If your package does not allow the max depth as a parameter, train trees with
2. Repeat point 1.) on the satimage, madelon, hill-valley, and covtype datasets.
3. On the Gisette dataset, for each off $k \in \{3, 10, 30, 100, 300\}$ train a random forest with k trees where the split attribute at each node is chosen from a random subset of $71 \approx \sqrt{5000}$ features. Use the trained trrees to predict the class labels on the training and test sets and obtain
4. Repeat point 3.) on the satimage, madelon, and hill-valley datasets.
5. Repeat point 3.) on the Gisette dataset where the split attribute at each node is chosen from a random subset of $12 \approx \log_2(5000)$ features, and where the split attribute at each node is chosen from all 5000 features, plotting all four curves (2 train and 2 test misclassification errors) on the same graph.